Scaffld: A Robot Teacher That Builds Its Own Lessons

One typed sentence → a new robot skill in 39.7 minutes

Team Scaffld · AgentVerse 2026 (UCL), Robotics Track · Technical summary, 8 November 2026 · github.com/owenbonc/scaffld

Abstract — Robots learn skills from human teleoperated demonstrations, which are slow and costly. **Scaffld removes the person:** a cloud teacher agent (Argon) turns one typed sentence into a checked task, demonstrates it by driving the robot, keeps only demos an independent physics validator accepts, and fine-tunes a small on-device student. On a Unitree G1 humanoid in MuJoCo, **one sentence taught the robot a new skill in 39.7 minutes** (wall-clock stage time): 56 accepted demos, a 21-minute fine-tune, and a 500M-parameter student that went from **0/20 to 18/20** on sealed, unseen scenes (**17/20** with the bin moved). Given a grey image instead of its camera, it scores **0/20** (on a Mac). It runs alone on a laptop GPU at **0.31 s per decision, with no cloud calls**.

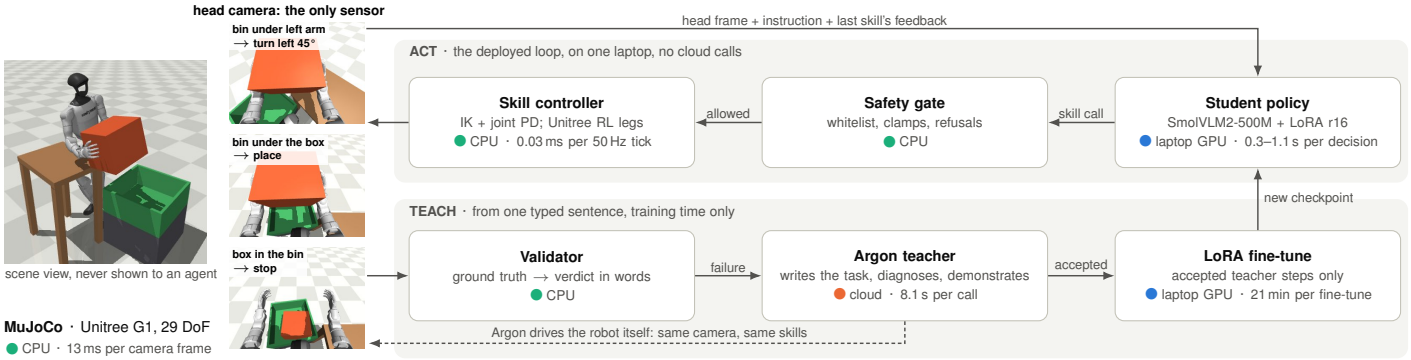


Figure 1. System architecture. **Act:** the deployed loop on one laptop; the head camera is the only scene sensor, and every skill call passes a safety gate. **Teach:** only the validator sees ground truth; Argon writes the task, diagnoses failures from saved frames and demonstrates through the same camera and skills; only accepted steps train the student. Frames: Argon's view after the bin moved, with its logged decisions. • laptop GPU • cloud • CPU.

1 User and problem

Users: robotics teams that must teach a robot a new task, or fix one it fails. **Problem:** that takes costly human demonstrations. DROID needed 50 collectors for 12 months (76k trajectories) [1]; RT-1 needed 13 robots for 17 months (130k episodes) [2]. **Our answer:** a teacher agent (Argon) that finds the gaps in a robot's skills, creates and validates its own lessons, and leaves a student that runs on the robot alone.

2 System architecture and key features

- **Robot and task.** A Unitree G1 (29 DoF) hugs a 0.8 kg box off a table into a bin, twisting at the waist (*fixed feet*) or walking 1.0–1.4 m on Unitree's RL policy (*walking*).
- **Camera in, skills out.** Agents see only a 320×240 head camera, the instruction and the last skill's feedback; ground truth reaches only the validator. They act through 10 whitelisted skills, run by IK and joint PD control.
- **Teaching (Fig. 1).** One typed sentence → Argon (gemini-4-argon) writes a task spec and success rule, checked in simulation → parallel Argon teachers demonstrate → the validator keeps what passes and names failures in words → accepted steps fine-tune the student → exam alone on 20 sealed seeds.
- **Key features:** mid-run box or bin moves that reach agents only as new frames; 13 worded verdicts; exact replay of accepted demos (56/56); a live dashboard where every number carries its n.

3 Where each model runs

Table 1. Measured on our hardware (laptop GPU: RTX 5070 Ti). Edge budget: ≤ 2 s per decision, ≤ 3 GB.

Model (role)	Runs on	Latency (n); memory
SmolVLM2-500M [3] + LoRA [4] (**student**)	• laptop GPU, bf16	0.31 s median, 0.56 s p95 (326); 2.01 GiB (v1/v2 exam)
LoRA r16 fine-tune (**training**)	• laptop GPU	21 min for 605 examples; 2.0 GiB
Argon, gemini-4-argon (**teacher**)	• cloud, teaching only	8.1 s median, 19.5 s p95 (3,551)
Unitree RL walking policy (**legs**)	• CPU, NumPy	0.03 ms per 50 Hz tick (2,000)
MuJoCo 4.0 + validator (**world**)	• CPU (Apple M5)	13 ms per camera frame (50)

4 Results

- **One sentence → new skill.** From "Stay where you are and put the box in the bin beside you, even if someone moves the bin", 8 parallel teachers made 60 demos in 16 min; the validator kept 56. After a 21-min fine-tune, the student went **0/20 → 18/20** on sealed scenes and **0/20 → 17/20** with the bin moved after the lift (Fig. 2a; McNemar $p < 0.001$, paired by seed). **39.7 min from sentence to tested skill** (wall clock, GPU queue excluded); a re-run reproduced every seed.
- **It acts on what it sees.** Given a grey image instead of its camera: **0/20 vs. 19/20** (same seeds, on a Mac). Blind, it twists once and places, so its decision to stop comes from the image.
- **Demos teach that decision.** Trained on only 14 or 28 of the same demos, it twists to its 150° limit and never places (0/20 each; Fig. 2b).
- **Lessons aimed at failures work.** Recovery and box-moved demos (52 → 117) raised success with the box moved from **7/20 to 15/20**, and on unseen bin placements from **14/40 to 21/40** (overall $p = 0.19$; bin further ahead 4/20 → 14/20, $p = 0.006$). In round 1, Argon named the cause of all 10 failed trials.
- **Inside the edge budget. 0.31 s median**, 0.56 s p95 ($n = 326$); 1.08 s over a 35-min exam ($n = 237$); 2 of its 2,145 decisions passed 2 s (the first after loading). Peak memory **2.01 GiB of 3 GB** (Fig. 2c).

5 Limits

Simulation only; sim-to-real is next. **One task family,** learned at the skill level. **Walking:** the memoryless student has not yet learned when to stop turning. **Bin moves** in our tests stay within reach of its learned routine; larger moves are next. **Teaching needs the cloud;** the deployed student does not.

6 Critical failure mode and safety

- **Failure mode: acting on a stale picture of the world.** If the box or bin moves mid-task, a robot replaying its plan grasps air or drops the box; near people, it swings a load into space that is no longer clear. **Our agents keep no plan:** every skill is chosen from the newest frame, and recovery lessons more than doubled success with the box moved (Fig. 2a).
- **When unsure, look again.** An unparseable answer becomes `reobserve`; two in a row end the episode.
- **Safe motion.** Only whitelisted, clamped skills run. A move past a limit, an unreachable target or a walk into the table is refused before anything moves. The validator rejects falls, joint-limit violations and penetration, so unsafe demos never train.
- **Data and secrets.** Sealed held-out seeds, an 800-call Argon cap per run, keys never logged; 600+ tests pin these rules.

[1] A. Khazatsky et al. DROID: a large-scale in-the-wild robot manipulation dataset. *arXiv:2403.12945*, 2024. [2] A. Brohan et al. RT-1: Robotics Transformer for real-world control at scale. *RSS*, 2023. [3] A. Marafioti et al. SmolVLM: redefining small and efficient multimodal models. *arXiv*, 2025. [4] E. J. Hu et al. LoRA: low-rank adaptation of large language models. *ICLR*, 2022.

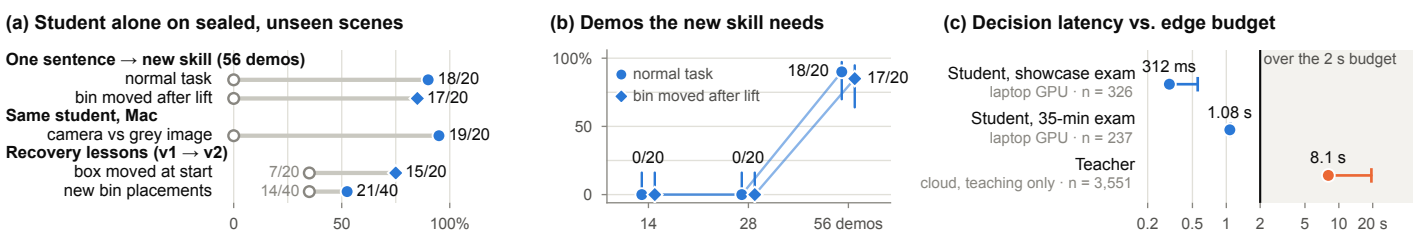


Figure 2. (a) Student alone on sealed scenes, each row against its control (hollow); diamond: scene changed mid-run. (b) First 14, 28 or 56 accepted demos; Wilson 95% CIs. (c) Dot median, tick p95.